# RBView User Manual

**LiDAR Point Cloud Visualization Software**

**Version V1.1.5**

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# Revision Information

This document is subject to ongoing revision based on user feedback and product updates without prior notice. Before using the product, visit the Richbeam official website at www.richbeam.com to obtain the latest version. If you have any questions about this document, contact us.

Feedback email: support@richbeam.com

## Revision History

| Date | Version | Description of Changes |
|---|---|---|
| 2022-06-15 | 1.0.1.0 | Initial release |
| 2022-08-25 | 1.0.1.1 | Updated software dependencies and added CSV export |
| 2024-10-16 | 1.0.5.0 | — |
| 2026-08-01 | 1.1.5.0 | Added Linux usage instructions and other updates |

# Overview

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This document describes the installation, features, and operation of Richbeam's LiDAR point cloud visualization software RBView and is continuously updated as the LakiBeam/LoraBeam product series evolves. RBView is intended solely for controlling and viewing data from LakiBeam/LoraBeam products and may not be used for any other purpose.

# 1 System Requirements

## 1.1 System Requirements

Operating system: Windows 10 (64-bit) or later, or Linux

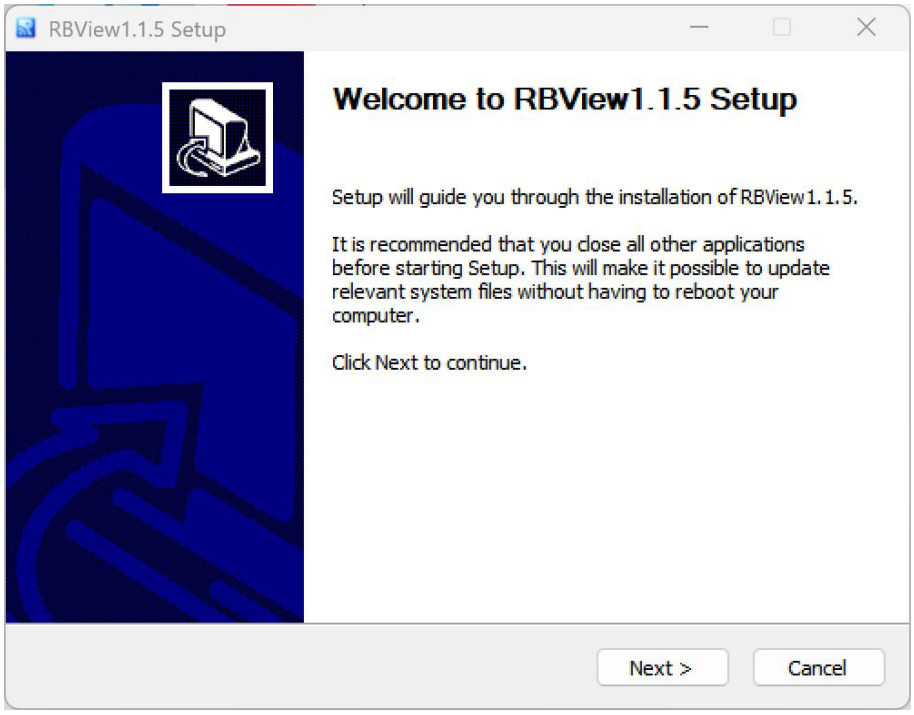
Memory: 4 GB or more

CPU: 4 cores, 2.0 GHz or higher

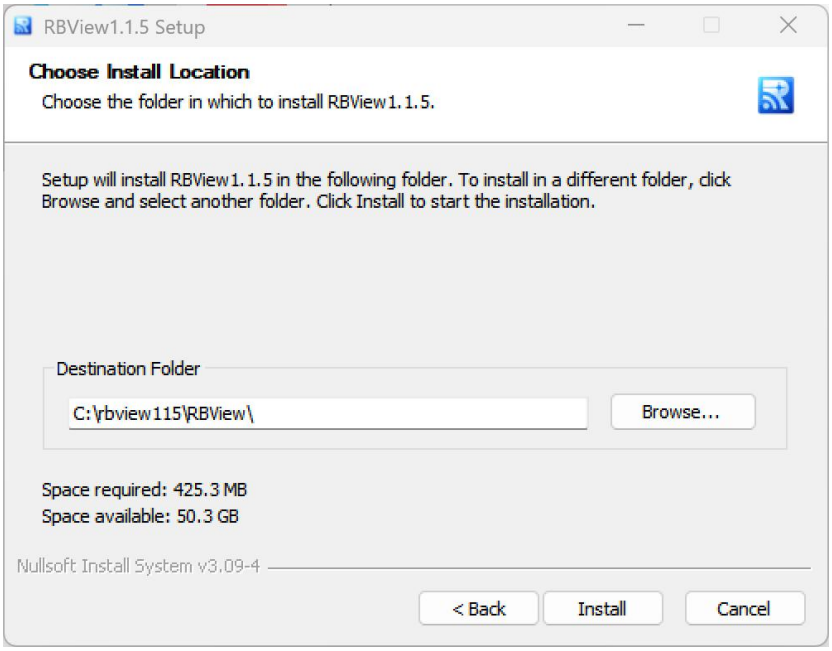
# 2 Installation

## 2.1 Installing on Windows

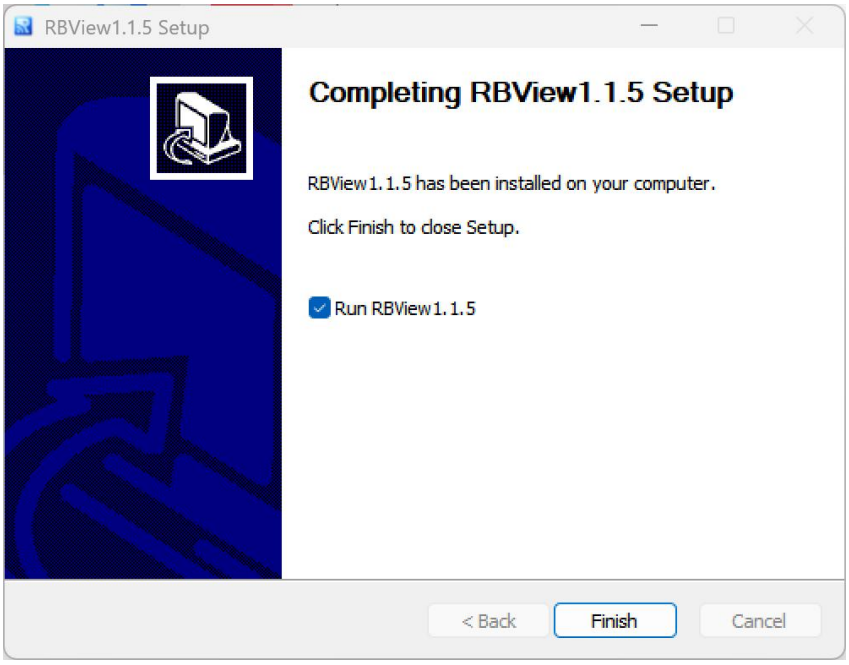
1. Double-click RBView-1.1.5-Setup-x64.exe to launch the installer, and then click "Next."



2. Select the installation folder, and then click "Install."




3. When installation is complete, click "Finish" to exit the installer. RBView is now ready to use.

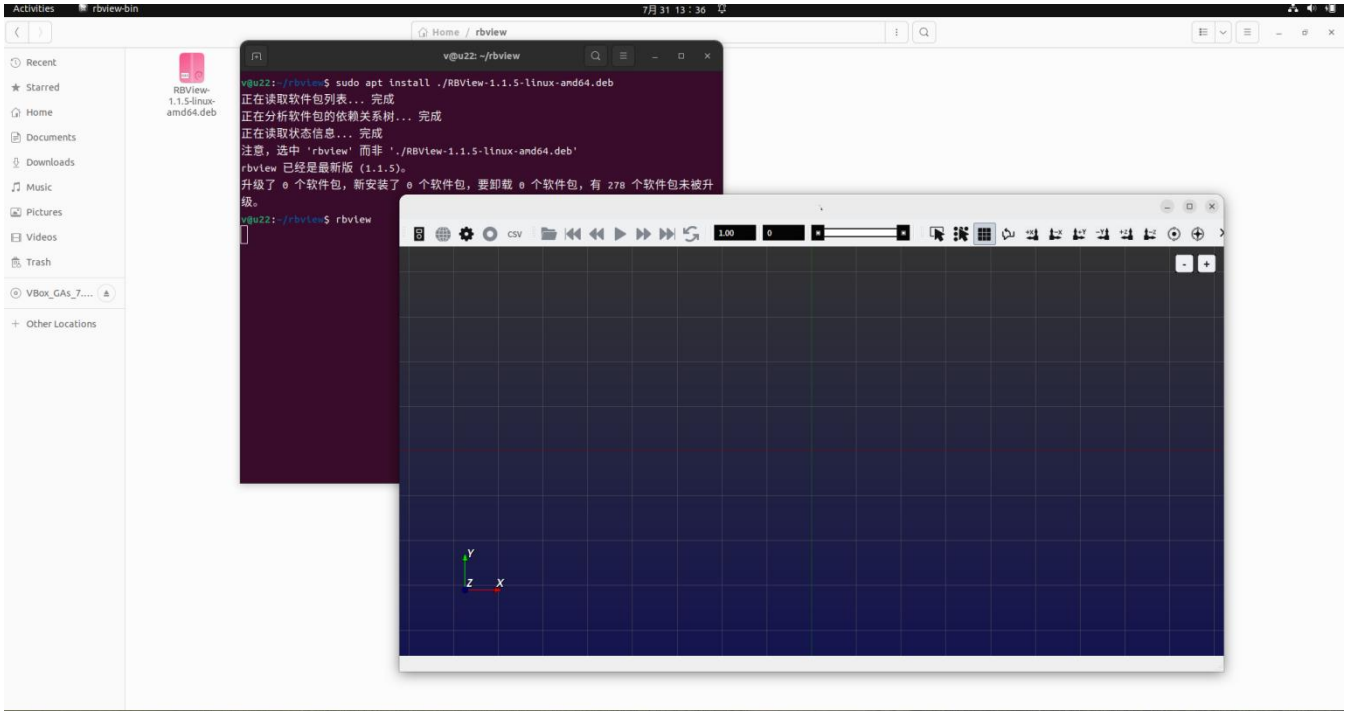


## 2.2 Installing on Linux

1. Run the following command in a terminal to install the DEB package:

```
sudo apt install ./RBView-1.1.5-linux-amd64.deb
```

2. After installation, enter rbview in a terminal to launch the application.



# 3 Network Configuration

## 3.1 Connecting the LiDAR

Connect the LiDAR to the power adapter and turn it on. Then connect the LiDAR to the computer using a 100 Mbps Ethernet cable. After power-on, the LiDAR performs a self-test for approximately 30 seconds. When the blue indicator flashes rapidly, the LiDAR is operating normally.

## 3.2 Windows Network Configuration

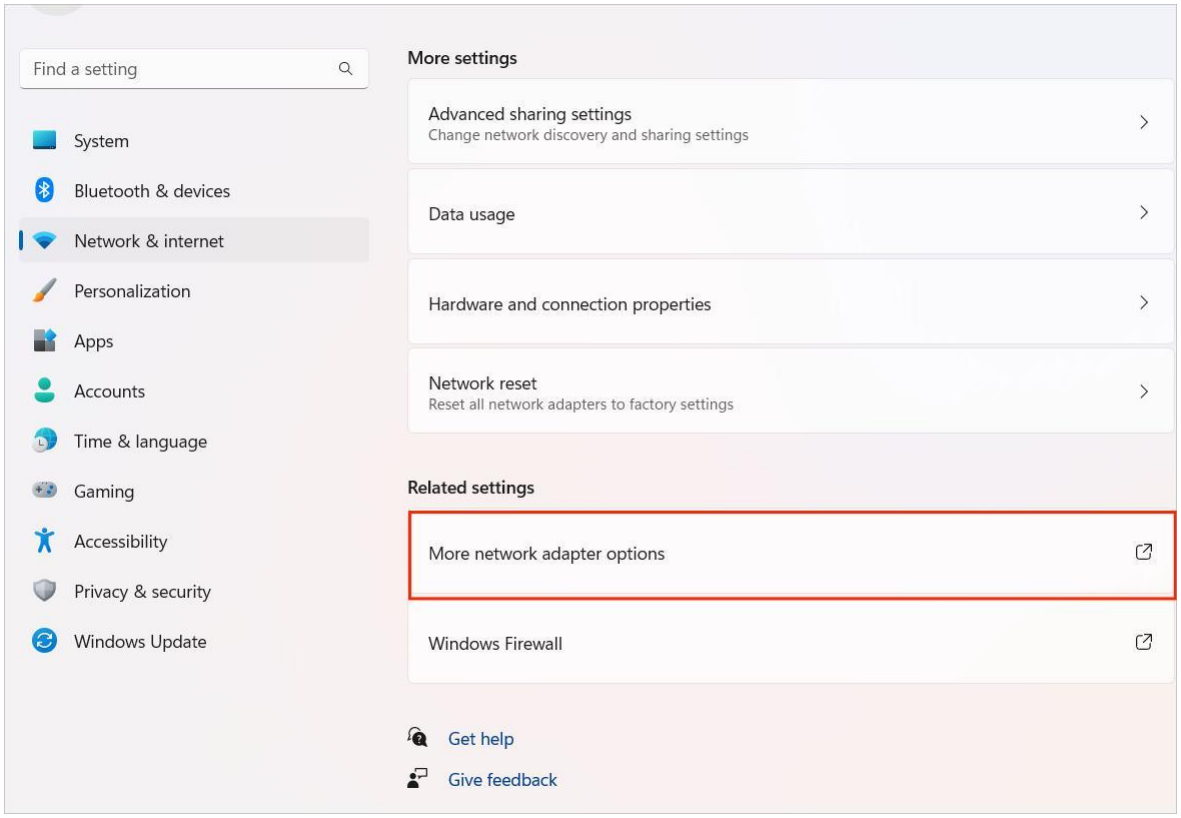
Before using RBView to view the point cloud data output by the LiDAR, configure the network interface used to connect the LiDAR to the computer.

After power-on, the LiDAR (LakiBeam/LoraBeam) automatically begins acquiring and transmitting data. By default, the LiDAR uses a static IP address of **192.168.198.2**, a subnet mask of **255.255.255.0**, and data port **2368**.

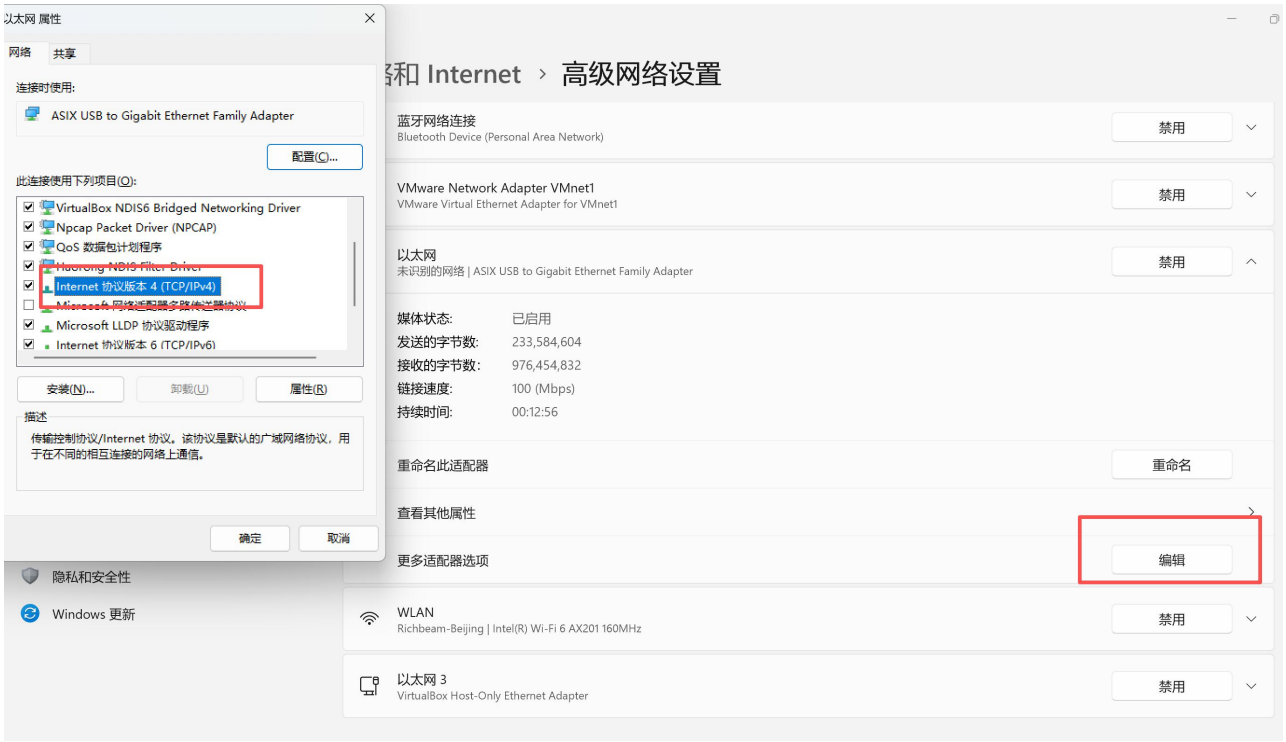
Therefore, set the IP address of the computer's network interface (or that of another device connected to the LiDAR) to **192.168.198.1**.

Follow these steps:

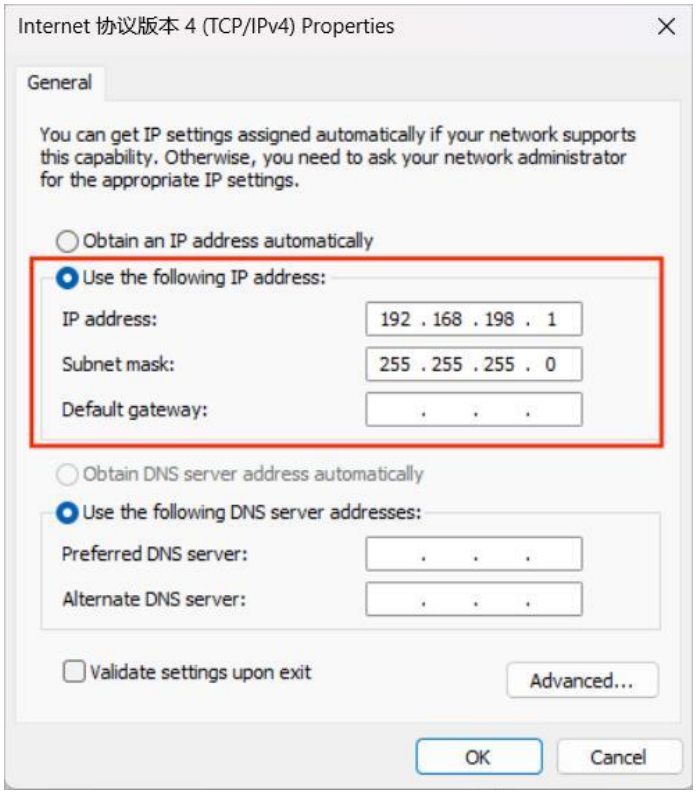
1. Open Windows "Settings," select "More network adapter options."



2. Locate the network adapter connected to the LiDAR, click "Edit," and then **double-click** "Internet Protocol Version 4 (TCP/IPv4)."

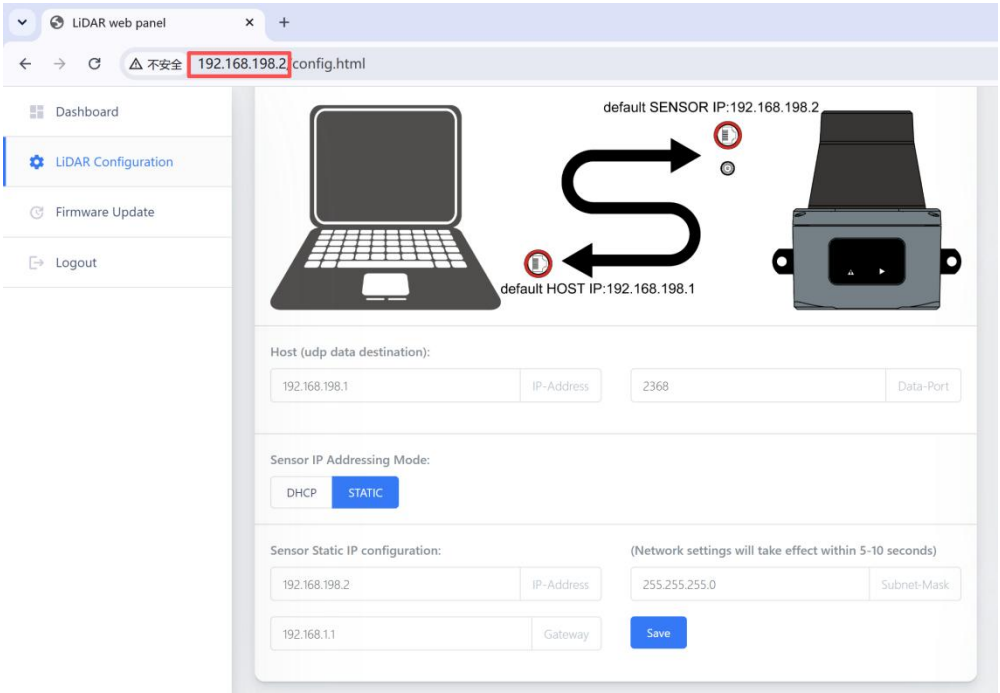


3. Select "Use the following IP address," set the IP address to **192.168.198.1** and the subnet mask to **255.255.255.0**, and then click "OK."



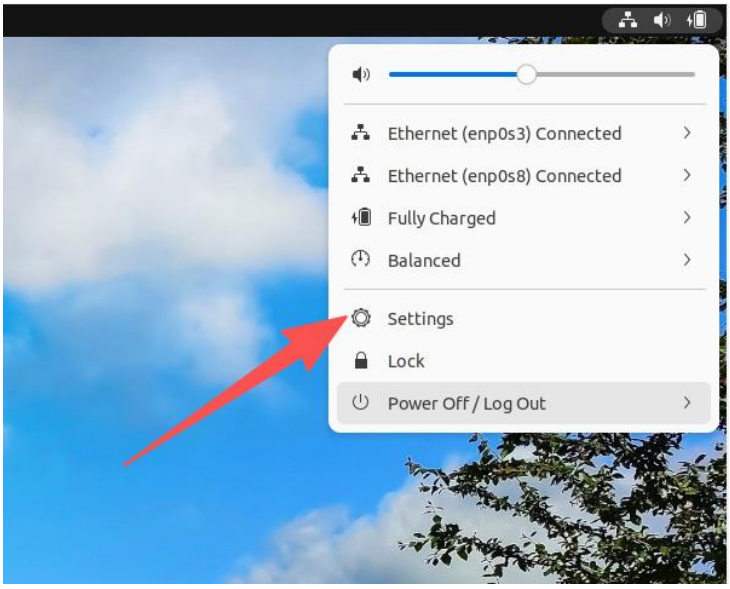

4. Open a web browser on the computer and enter 192.168.198.2 in the address bar.

If the LiDAR web interface opens, the connection has been established successfully.

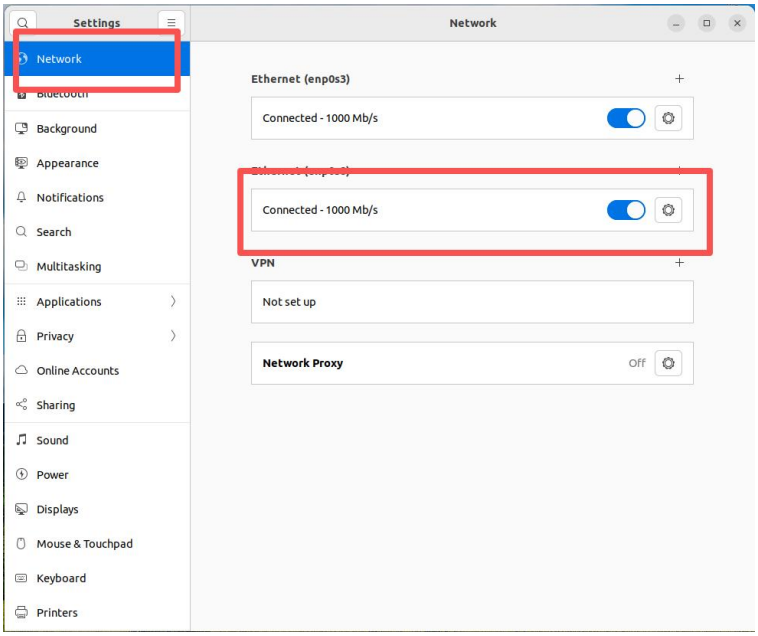


## 3.3 Linux Network Configuration

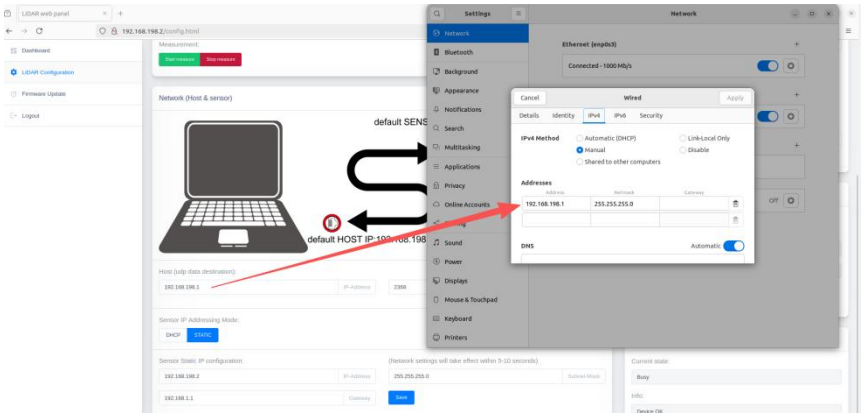
1. Open the system "Settings."




2. Locate the network interface connected to the LiDAR, and then click the settings button.



3. Set "IPv4 Method" to "Manual," set the network interface IP address to **192.168.198.1**, and set the subnet mask to **255.255.255.0**. This address must match the Host IP setting in the LiDAR web interface.

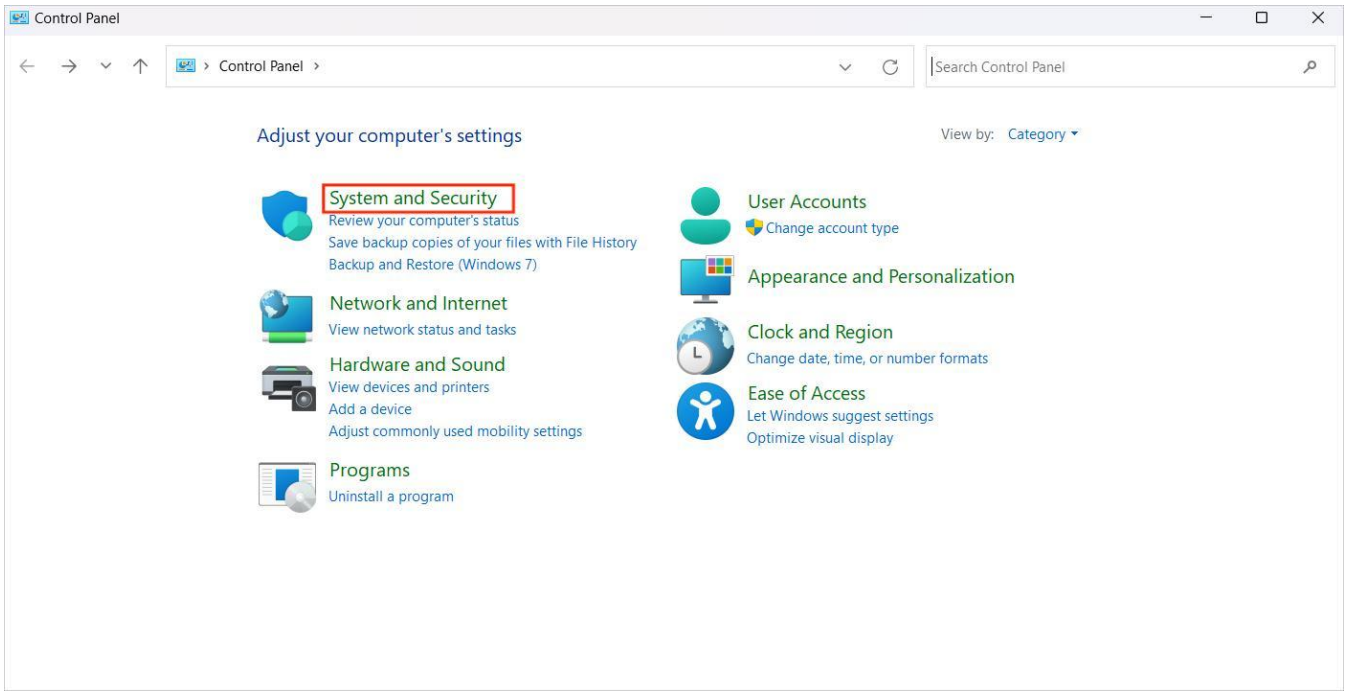


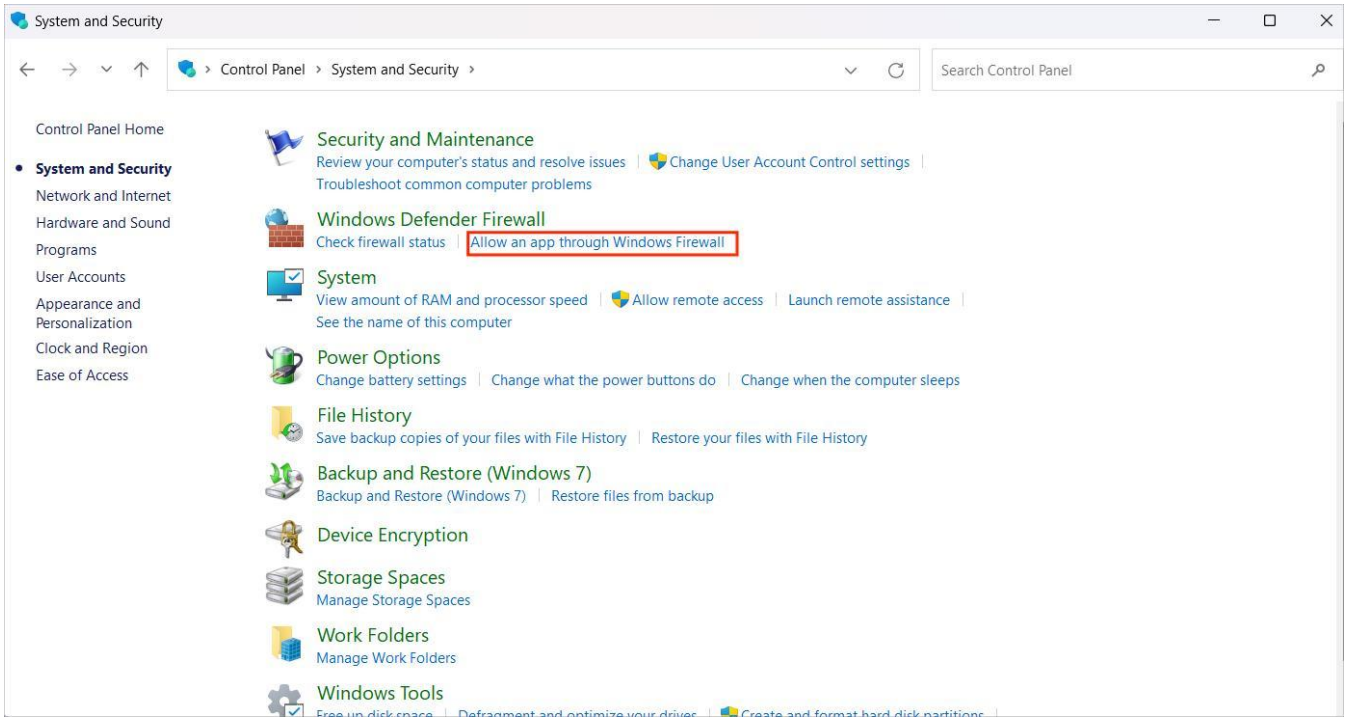
# 4 Firewall Configuration

## 4.1 Windows Firewall Configuration

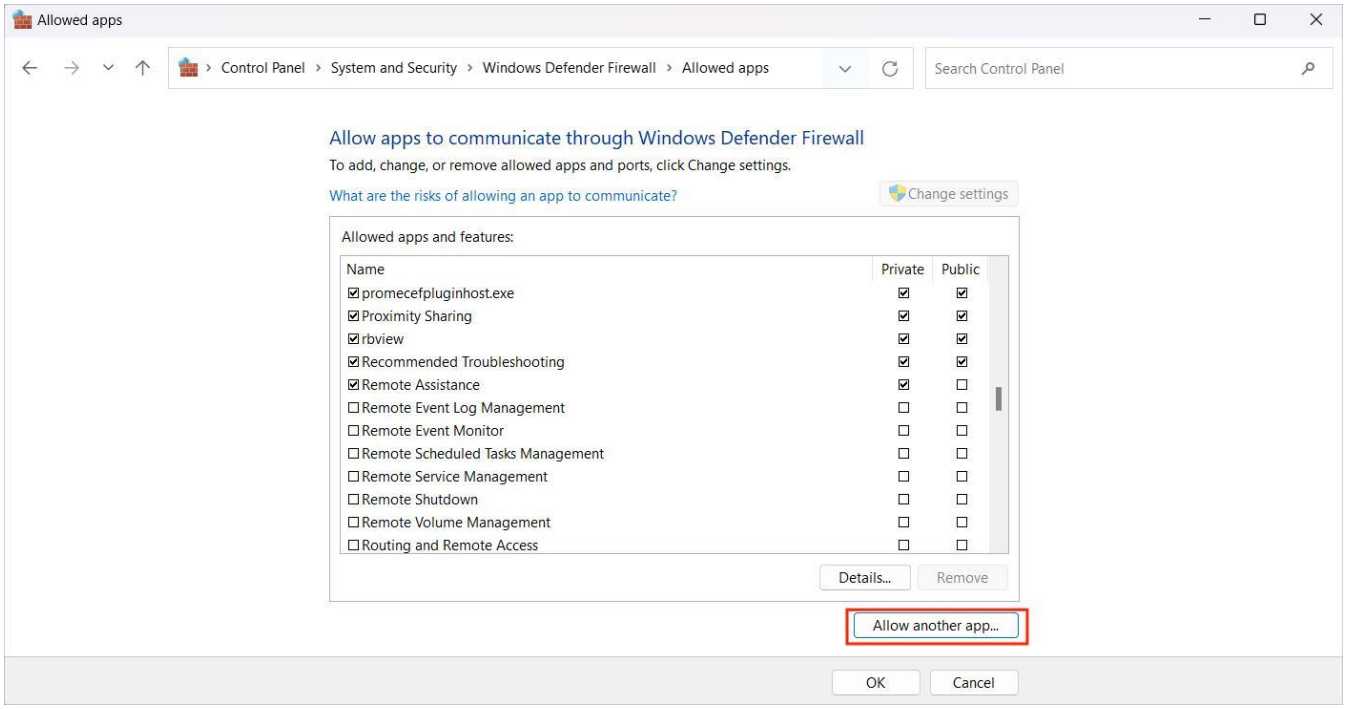
The LiDAR continuously sends UDP packets to the computer through the network interface. If the operating system firewall blocks the packets, RBView cannot receive them correctly. Before use, configure a firewall rule that allows RBView to communicate.

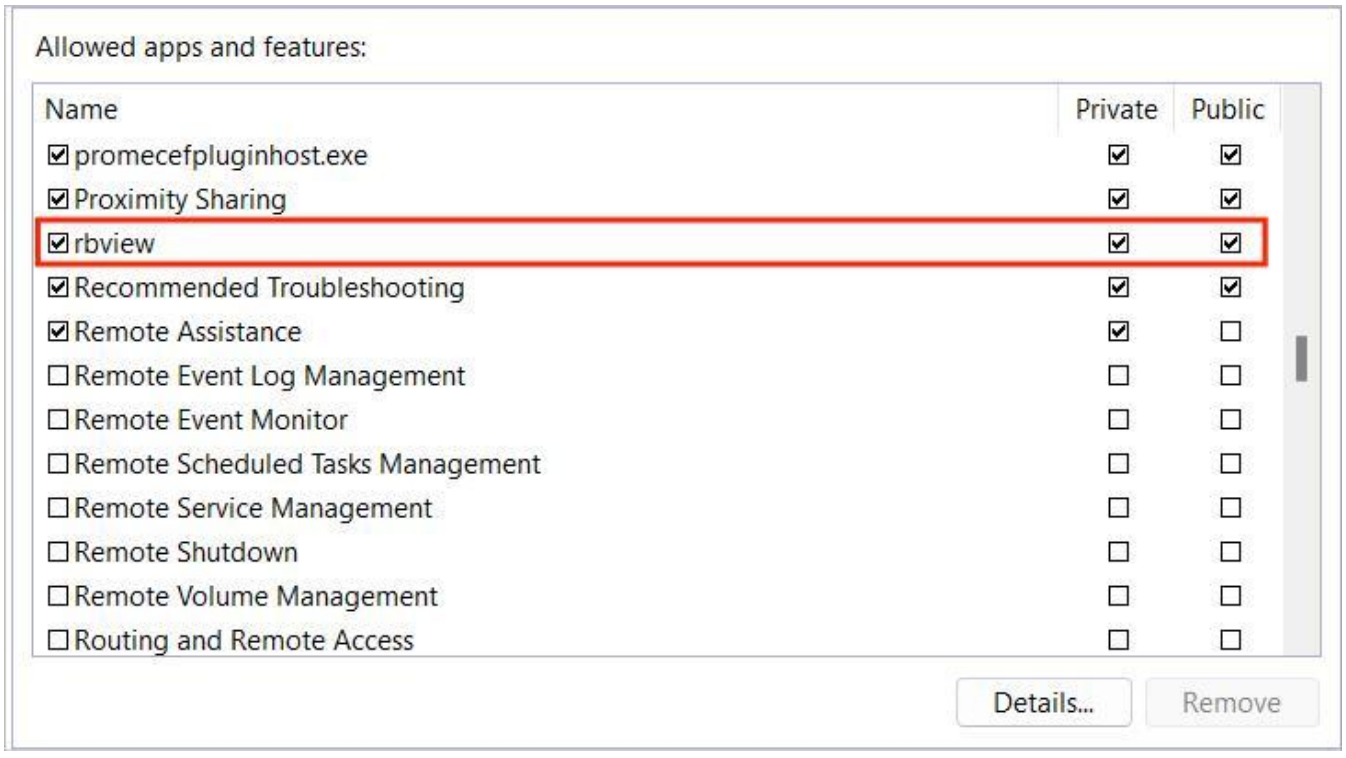
1. Open "Control Panel," select "System and Security," and then select "Allow an app through Windows Firewall."



2. Click "Change settings," add RBView.exe, and allow RBView to communicate through Windows Firewall. When finished, click "OK."



## 4.2 Linux Firewall Configuration

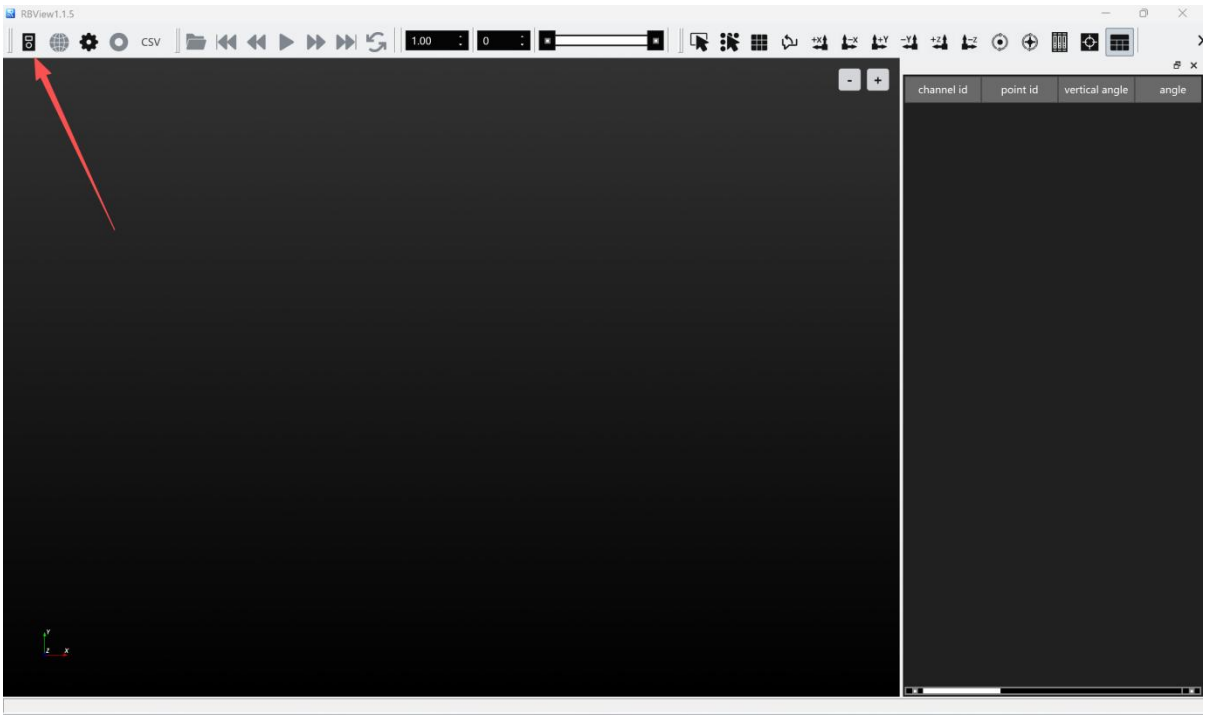
Enter **sudo ufw disable** in a terminal.

When using LakiBeam on Linux, **keep the firewall disabled at all times** to prevent abnormal point cloud data or a complete loss of point cloud data.

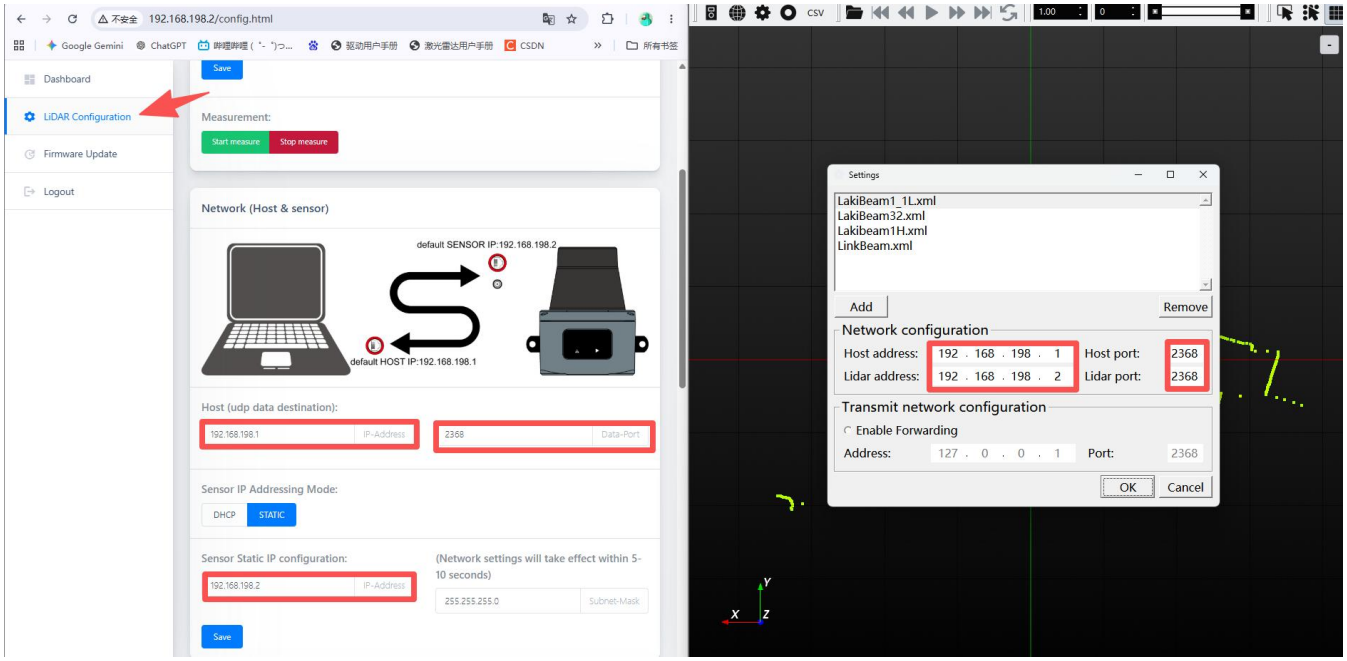
# 5 Using RBView

## 5.1 Acquiring LiDAR Data

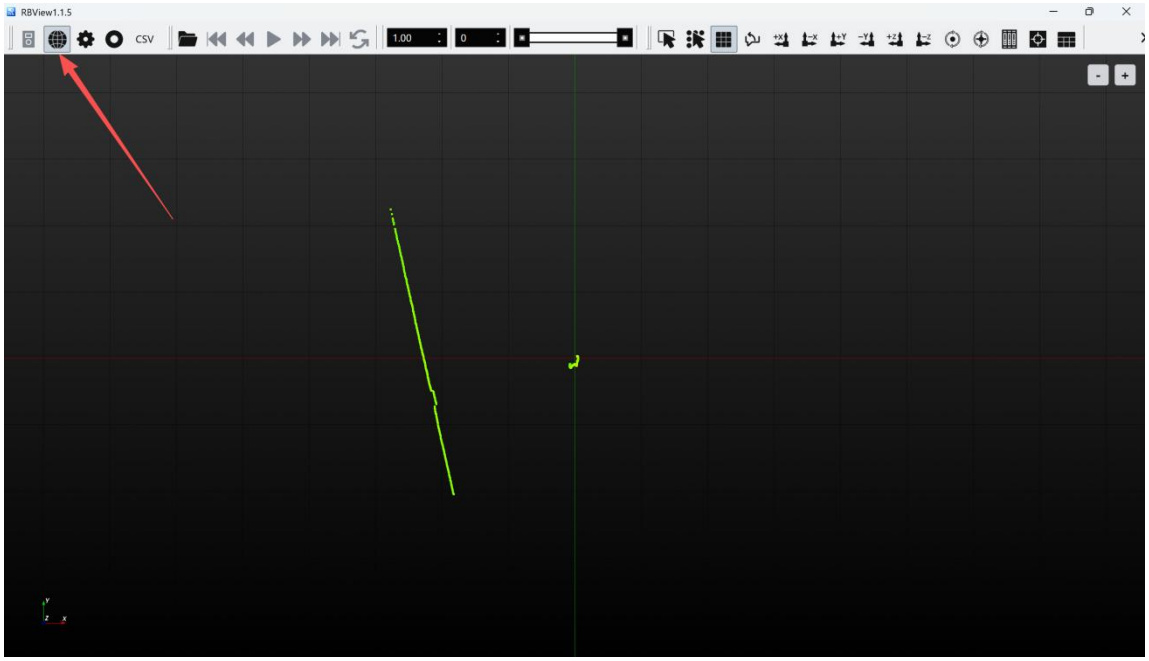
1. Open RBView and click the "Open lidar configurations" icon.




2. In the "Settings" window, select LakiBeam1_1L.xml. Confirm that the LiDAR IP and Host IP under "Network configuration" match the settings in the LiDAR web interface, and then click "OK."



3. Click the "network"  icon to view point cloud data. Click the icon again to pause acquisition.




4. Basic point cloud view operations are as follows:

Hold down the left mouse button and drag: Rotate the coordinate system.

Hold down the right mouse button and drag up or down, or scroll the mouse wheel: Zoom the coordinate system and the entire point cloud.

Hold down the mouse wheel and drag: Pan the coordinate system and the entire point cloud.

Click the plus/minus icons 

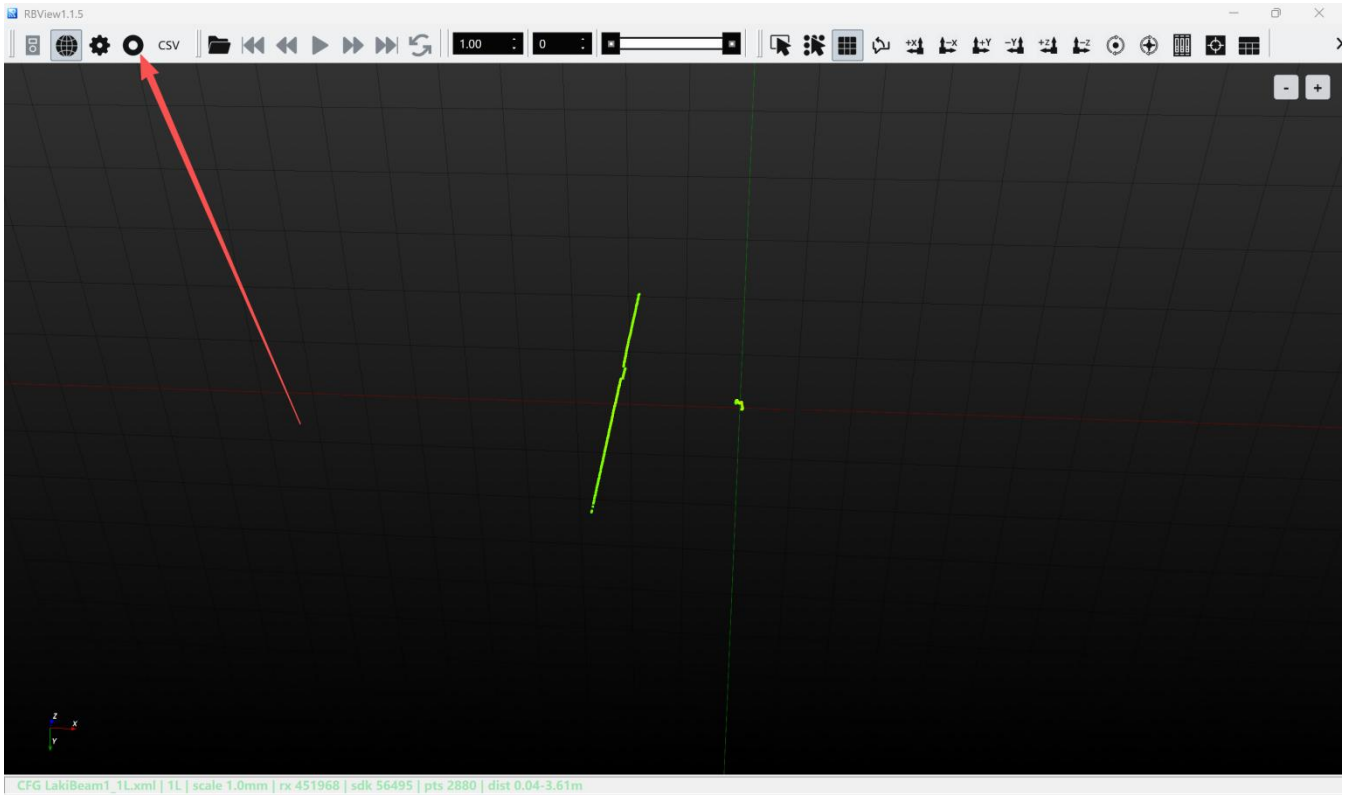
: Adjust the displayed point size in the point cloud.

## 5.2 Recording Point Cloud Data

While acquiring point cloud data, RBView can record and save the live data for later playback and analysis.

Follow these steps:

1. Click the record button .




2. Select a save location and enter a file name.

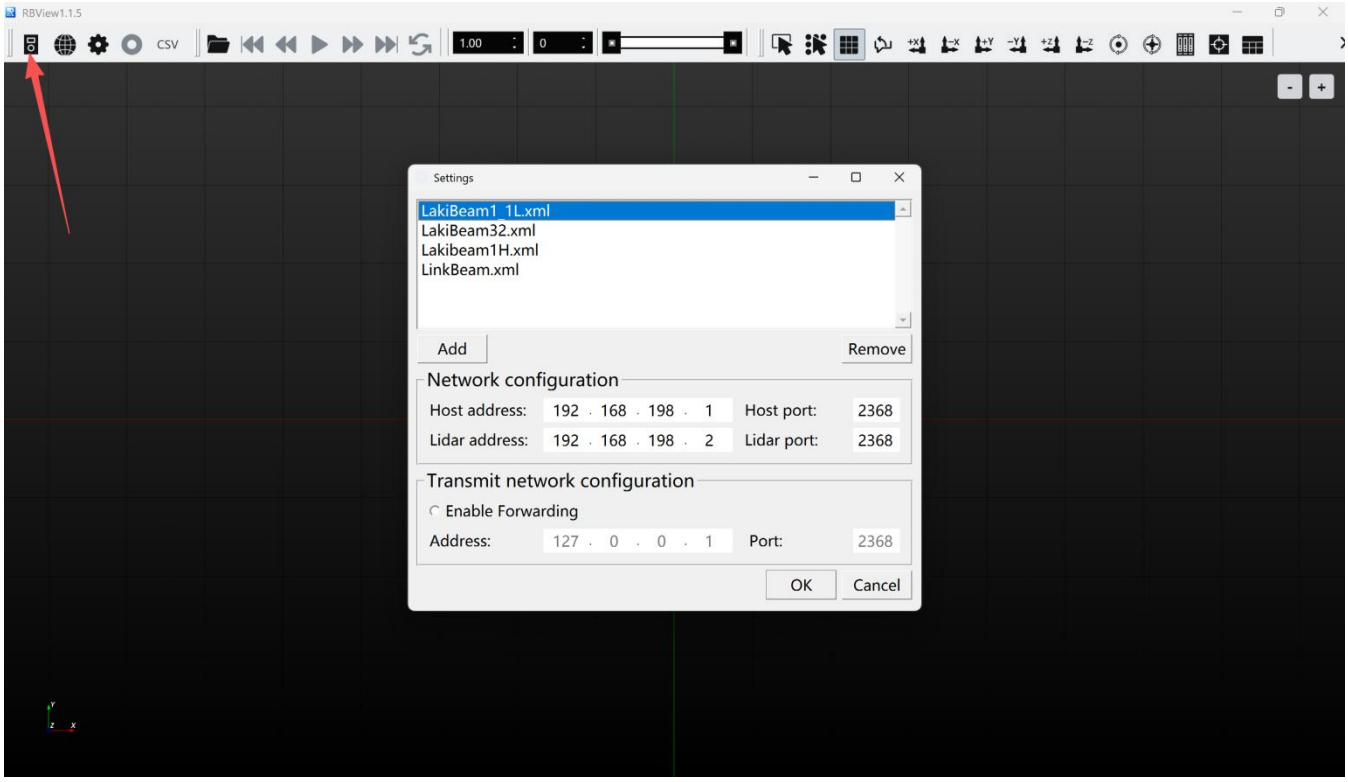
3. Click "Save" to begin recording.

4. Click the record button again to stop recording.

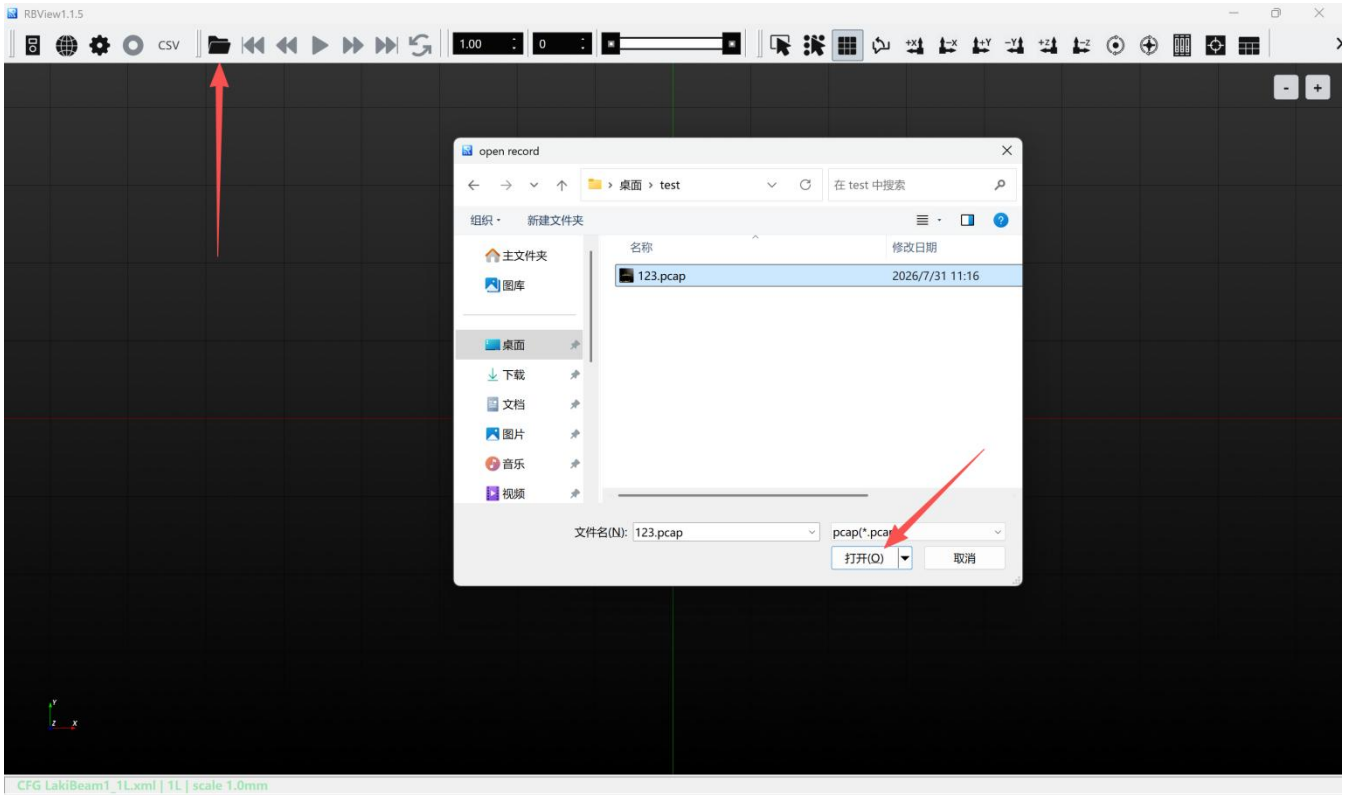
## 5.3 Playing Point Cloud Data

After recording, load and play the point cloud data in RBView for further analysis. Follow these steps:

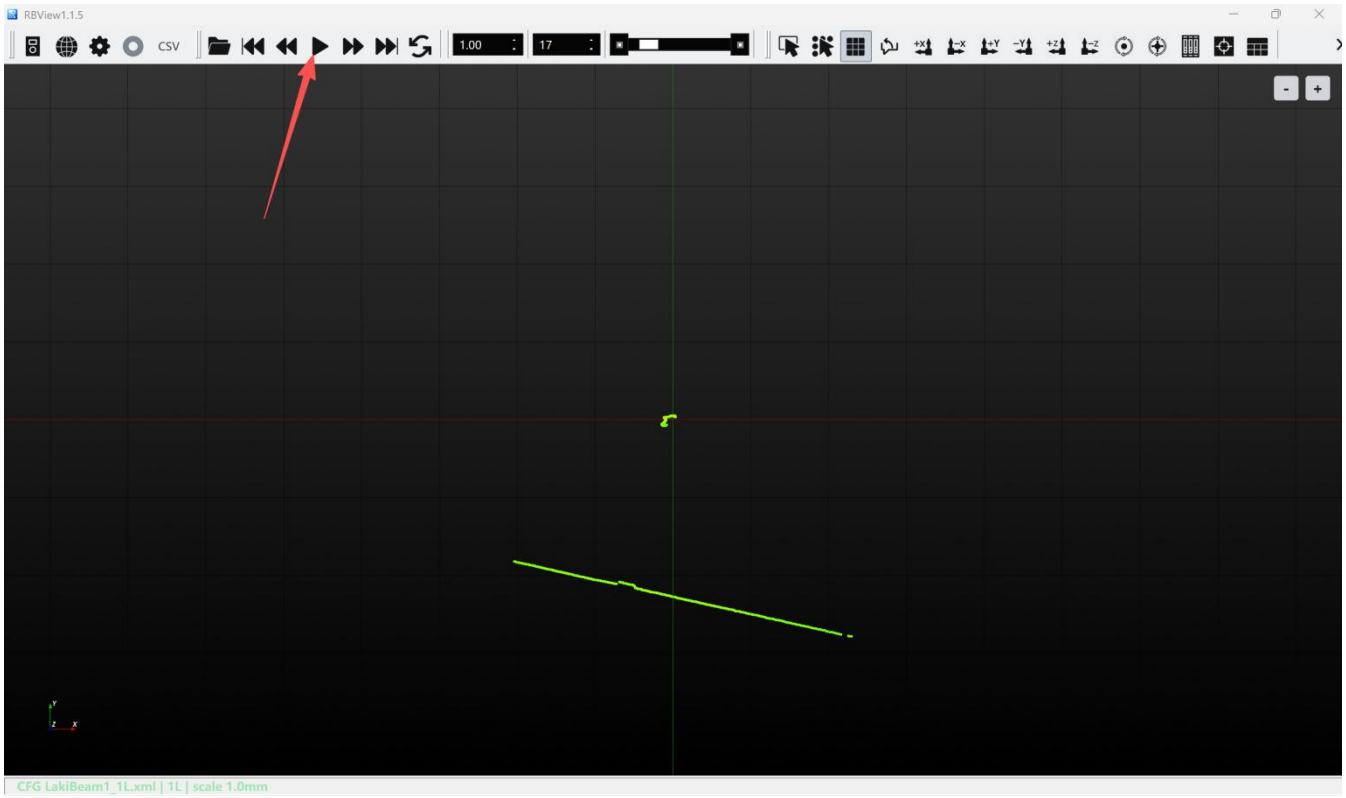
1. Click the "Open lidar configurations"  icon. Configure the options in "Settings," and then click "OK."




2. Click the "Open record file"  icon, select a recorded data file, and then click "Open."



3. Click the "Play" button  to begin playback.




From left to right, the buttons shown below are: first frame, previous frame, play/pause, next frame, last frame, and loop playback.



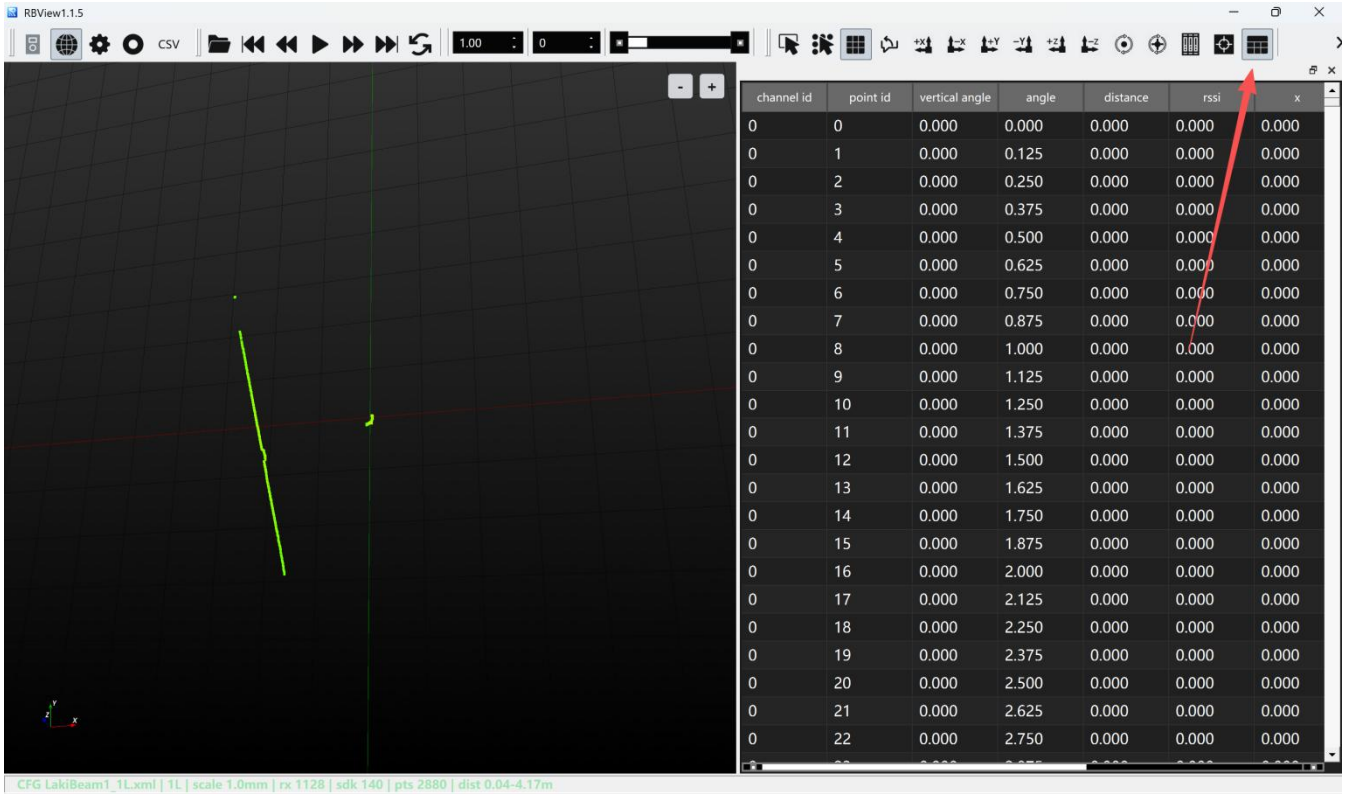
The point cloud playback progress bar is shown below:



## 5.4 Viewing Detailed Data

RBView provides a data table in the side panel for viewing detailed information about any point. Click the table button  to open a table containing all point cloud information, as shown below:

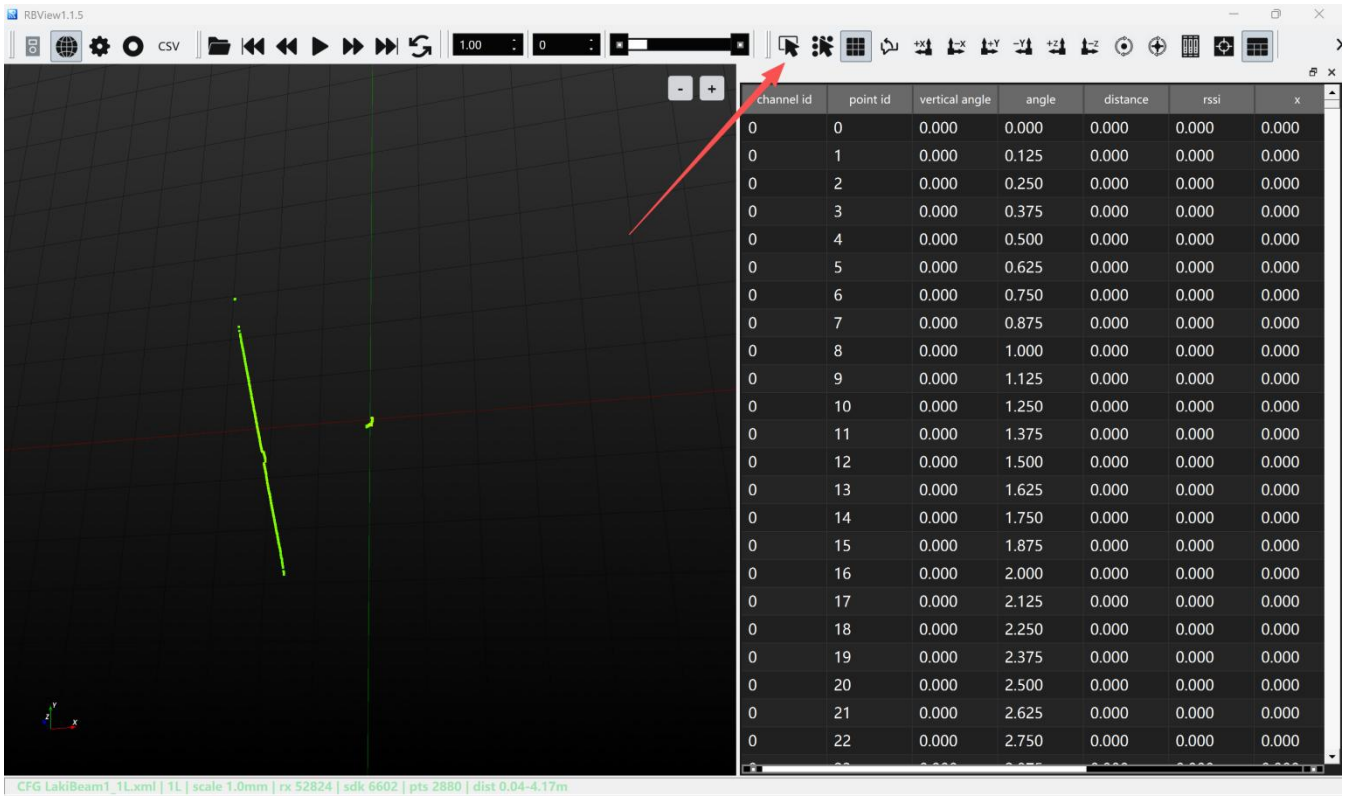
The table includes the channel ID, point ID, XYZ coordinates in the Cartesian coordinate system, horizontal angle, target distance, signal intensity, and timestamp.



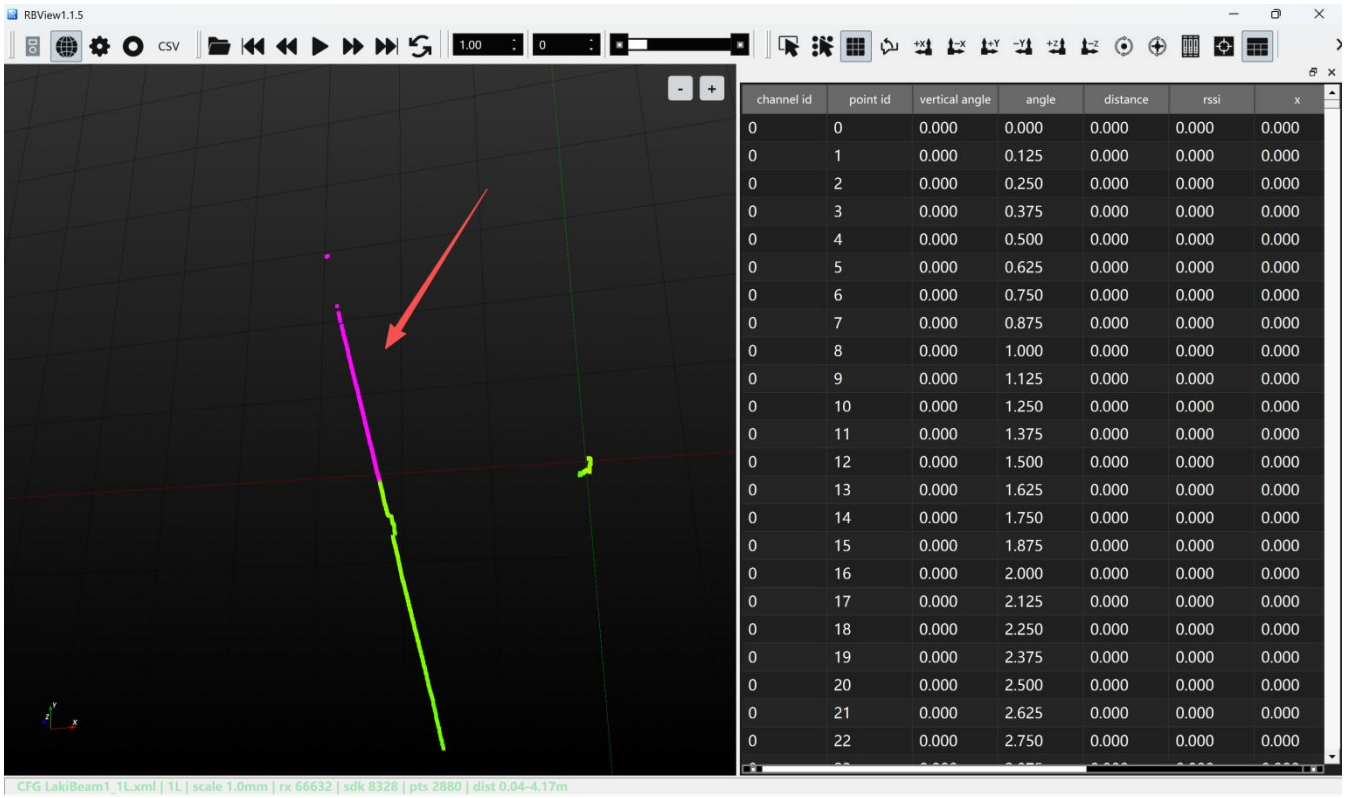
## 5.5 Region Selection

RBView provides a region selection feature for viewing point cloud information within a specified region while filtering out other points. Follow these steps:

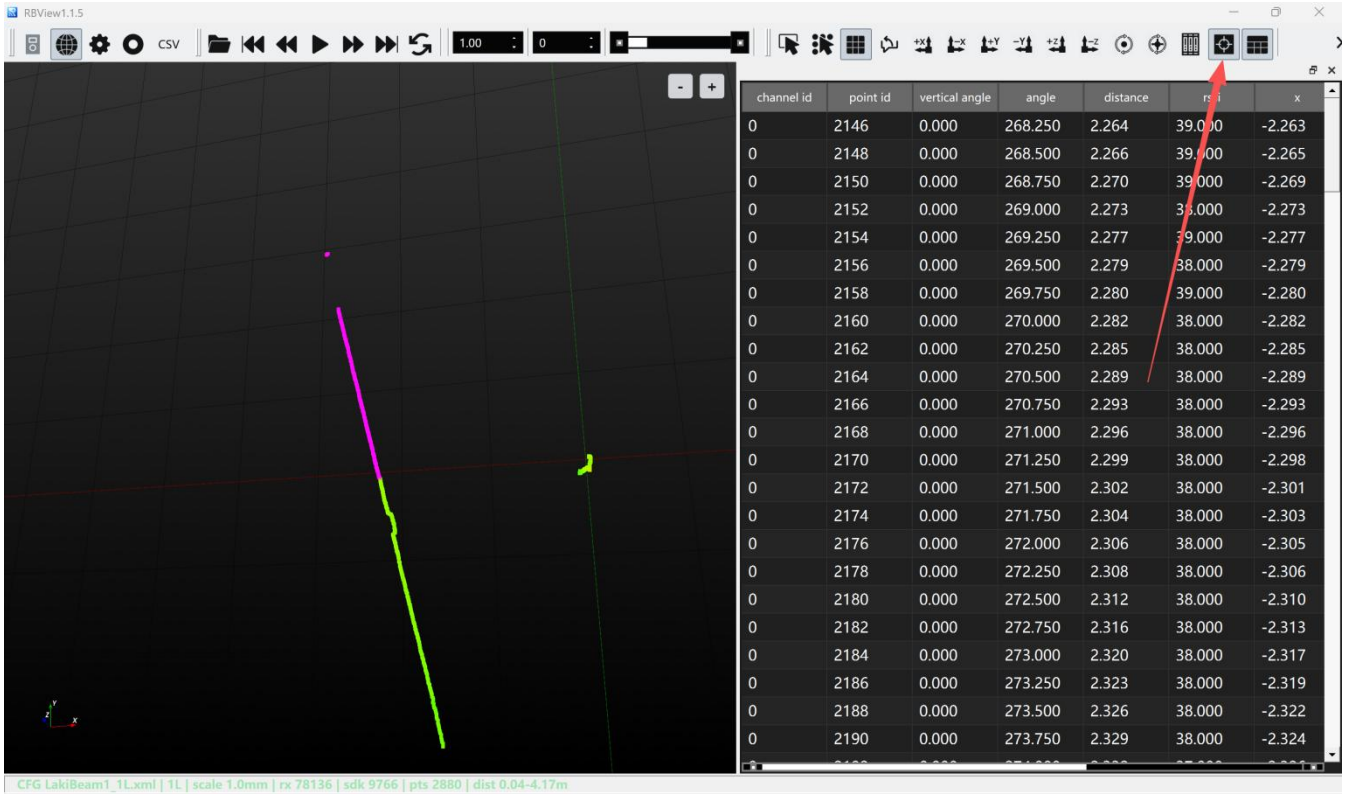
1. Click the region selection button  .




2. Drag a selection box around the point cloud region to be viewed.



3. Open the side data table, and then click the "tables show select" button  .




The data in the selected region is highlighted in purple and displayed in the side-panel table. To clear the selection, click the  button on the toolbar.



## 5.6 Exporting CSV Files

RBView can export point cloud data to a CSV file. After clicking the "save as csv" button, use the frame selection window to select the current frame (pause acquisition first), all frames, or a specified frame range, and then save the data to the selected folder. Follow these steps:

1. Click the pause button  to pause point cloud acquisition.

2. Click the "save as csv" button CSV.

3. Select the frame range to export, and then click "OK."

